INCH-POUND

MIL-PRF-52343D
15 September 1998
SUPERSEDING
MIL-R-52343C
12 September 1988

PERFORMANCE SPECIFICATION

REPAIR KIT, HOSELINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a repair kit consisting of tools and materials necessary for temporary and permanent repairs to 4-inch hoselines. The components are contained in a fitted storage chest. The kit is not intended for making repairs to potable water hoseline systems.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-D/210, Warren, MI 48397–5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3835

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SPECIFICATIONS

FEDERAL

A-A-1399	– Socket, Socket Wrench, Nonpower, ½ Inch Drive, 12-Point, Regular Length
A-A-1689	– Tape, Pressure-Sensitive Adhesive (Plastic Film)
A-A-2166	– Handle, Speeder, Brace Type
A-A-2476	– Hammer, Hand (Nonsparking and Nonmagnetic)
A-A-2522	– Rags, Wiping (Cotton and Cotton-Synthetic)
A-A-50271	– Plate, Identification
A-A-52598	– Couplings, Clamp, Pipe; with Bolts and Synthetic-Rubber Gaskets for Grooved-End Pipe

STANDARDS

FEDERAL

MIL-STD-130	– Identification Marking of U.S. Military Property

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents that are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 580	– Standard Specification for Anodic Oxide Coatings on Aluminum
ASTM D 235	– Standard Specification for Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent)
ASTM D 3953	– Standard Specification for Strapping, Flat Steel and Seals
ASTM D 4388	– Standard Specification for Nonmetallic Semi-Conducting and Electrically Insulating Rubber Tapes

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ASME/ANSI B107.15	– Flat Tip and Phillips Screwdrivers
ASME/ANSI B1.5	– Acme Screw Threads
ANSI B1.9	– Buttress Inch Screw Threads 7°/45 Form with 0.6 Pitch Basic Height of Thread Engagement

(Application for copies should be addressed to the American National Standard Institute, 11 W. 42nd Street, New York, NY 10036.)

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2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The hoseline repair kit (hereinafter called "kit") shall consist of the components described herein and be fitted in a storage chest.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Repair kit parts list. The components comprising the repair kit and their required quantities are listed in table I.

TABLE I. Hoseline Repair Kit Components.

Qty.	Description	Reference Number
10	Grooved-end adapters	Figure 1
10	Coupling, 4 inch NPS	A-A-52598
5	Rubber adhesive (8-oz.) with brush applicator	(see 3.5)
4	Hose clamp	(see 3.6)
1	Hammer, Type I, ball peen, machinist, 32 ± 4 oz.	A-A-2476
1	Knife, produce or square point, 6-inch blade	
150	Rubber tape (feet), Type III, 1 inch wide × 0.030 inch thick	ASTM D 4388
350	Vinyl-plastic tape (feet), Type I, Class I	A-A-1689
1	Banding tool, manual, 3/4 inch	(see 3.7)
2	Banding strap, Type I, heavy duty, Finish B, Grade 1, 3/4 inch wide × 0.030 inch thk (100 ft. roll)	ASTM D 3953
100	Banding strap seals, Style I, 3/4 inch size	ASTM D 3953
3	Solvent (1-quart), Type III, Class A	ASTM D 235
1	Screwdriver, Type I, Class 2, Design A, 4 pt. size	ASME/ANSI B107.15
1	Socket wrench, 1/2 inch drive 12 point, 1-1/16 inch, non-power	A-A-1399
1	Handle, speeder, brace type, 1/2 inch drive	A-A-2166
50	Rags (pound) P/N AA2522-A1 (see 3.8)	A-A-2522
2	Hex key, T-handle, 3/8 inch, 12 inch long	
1	Storage Chest	(see 3.8)

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3.4 Fittings.

3.4.1 Grooved-end adapters. Each adapter shall be in accordance with the dimensions shown in figure 1. Each adapter shall be anodized in accordance with ASTM B 580, type A.

3.4.2 Coupling. The grooved-end pipe coupling shall be a 4-inch grooved-end pipe coupling in accordance with A-A-52598, part identification number AA52598-A-1143.

3.5 Rubber adhesive. Rubber adhesive shall be supplied in leakproof cans. Each can shall have an adhesive applicator attached to the top suitable for applying the adhesive. The rubber adhesive shall be the cold-bond type that shall be used for bonding synthetic rubber to synthetic rubber.

3.6 Hose clamp. The hose clamp shall consist of a base, fastener, frame, crosshead, a riser, and riser handle or other means for lowering the crosshead to compress the hose (see figure 2).

3.6.1 Base. Means shall be provided for latching and pivoting the frame to facilitate encompassing the hose. The face of the lower jaw shall be a curved segment and shall have a concave cross section.

3.6.2 Fastener. The fastener shall be self-actuating and affixed to the base in such a manner that it moves freely. Means shall be provided for quickly engaging and locking the free side of the frame to the base.

3.6.3 Frame. The frame shall be U-shaped. One leg of the frame shall be hinged to the base and the other shall have the capability of being securely engaged by a fastener. The inside of the frame shall have parallel tracks to act as guide for the crosshead. The crosshead shall slide vertically over the frame without binding. The frame shall have a minimum 7.75-inch clearance between the legs of the “U” and shall provide not less than 4.50-inch clearance between the lower jaw and the crosshead when the frame is closed and the crosshead is in its uppermost position.

3.6.4 Crosshead. The lower face of the crosshead shall be a curved segment and shall have a convex cross section. The top of the crosshead shall incorporate a method for quickly lowering the crosshead so that a hose can be compressed against the lower jaw and stop the flow of fluid through the hose without damaging the hose.

3.6.5 Riser. If a screw is selected for positioning the crosshead it shall be made from 0.875-inch (minimum) diameter stock. The screw shall have American National Square Acme, or Modified Acme threads conforming to ASME/ANSI B1.5 or buttress threads (class 1A or 1B) conforming to ANSI B1.9. One end of the screw shall have means to receive the riser handle. The assembly shall be designed to prevent pinching the hands of the operator. The other end shall allow the screw to swivel in the crosshead. The screw shall be coated and finished per the manufacturer’s standard practice.

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3.6.6 Riser handle. The handle shall slide in the crosshead riser without binding to facilitate torquing, and shall be equipped at each end with a means to prevent the handle from becoming detached.

3.7 Banding tool. The manual banding tool shall accommodate a 3/4-inch (nominal) wide band. The tool shall be designed to apply tension to the strapping band when in place on the hose, and shall hold the tension while the banding seal is being fastened. The tension shall be applied to the strapping band by a screw or other suitable method. The tool shall not damage the hose while tension is applied to the band or while the seals are being fastened.

3.8 Storage container. All components of the repair kit shall be fitted in a storage container. The container shall have a hinged cover, fastening devices, and handles. Bins, cradles, or blocking for components of the repair kit shall be provided to prevent damage during shipment or handling. The container shall not exceed 22 inches in height. A loading plan indicating the location of all items in the container shall be provided. The loading plan shall be permanently attached to the inside of the cover. Space for at least 3 pounds of rags shall be provided.

3.9 Marking. The repair kit shall be identified and shall have warning labels in accordance with A-A-50271. The warning shall state “KIT NOT TO BE USED FOR REPAIRING POTABLE WATER HOSES”. Marking shall be applied to the outside cover of the storage container.

3.9.1 Identification markings. Identification markings shall be permanent, legible, and shall include, as a minimum per MIL-STD-130, the manufacturer’s identification code, the manufacturer’s part number the part or identification number (PIN), and the national stock number (NSN).

3.10 Materials. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. The materials shall be of sufficient durability to meet all the performance requirements as specified herein. The material shall have no adverse effect on the health of the personnel when used for its intended purpose. Whenever feasible, toxic chemicals, hazardous substances, or ozone depleting chemicals (ODC) shall not be used.

3.10.1 Deterioration prevention and control. The equipment shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration to which the materials are susceptible.

3.10.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion (See 6.5.1).

3.10.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Used, rebuilt or re-manufactured parts shall not be used in the repair kit.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article inspection is required, it shall be performed on one complete unit. The inspection shall include the examination 4.4.

4.3 Conformance inspection. Conformance inspection shall include the examination 4.4.

4.4 Examination. Each repair kit shall be physically examined for compliance with the requirements specified in table II. Any redesign or modification to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all examinations of performance, safety, human engineering, and dimensional requirements. Non-compliance with any specified requirement, or the presence of one or more defects shall constitute cause for rejection.

TABLE II. Examination.

Number	Examination Description	Requirement Paragraph
101	Repair kit components not as specified.	TABLE I
102	Storage chest not as specified.	3.8
103	Marking not as specified.	3.9
104	Identification not as specified.	3.9.1
105	Materials not as specified.	3.10
106	Deterioration prevention and control not as specified.	3.10.1
107	Dissimilar metals not protected against galvanic corrosion as specified.	3.10.1.1
108	Used, rebuilt or re-manufactured parts used.	3.10.2

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The kit is intended for use in making repairs on the 4-inch, lightweight, collapsible petroleum hoseline system. The hose clamps are used to crimp pressurized hoses on both sides of a damaged section and enable the repair crew to cut the damaged section out and install hose couplings or repair assemblies contained in the kit to effect a repair with minimum time and effort. This system is military unique in that no commercial equivalent exists for repairing hose of this size and type, or for withstanding repairs under battlefield conditions. The kit is not intended for making repairs to potable water hoseline systems.

6.2 Acquisition requirements. Acquisition documents will specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
c. When first article is not required or time frame for submission of first article (see 3.2).
d. Packaging requirements (see 5.1).

6.3 First article inspection. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article following testing.

6.4 Data Requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, and parts list to be furnished with each repair kit.

6.5 Definitions.

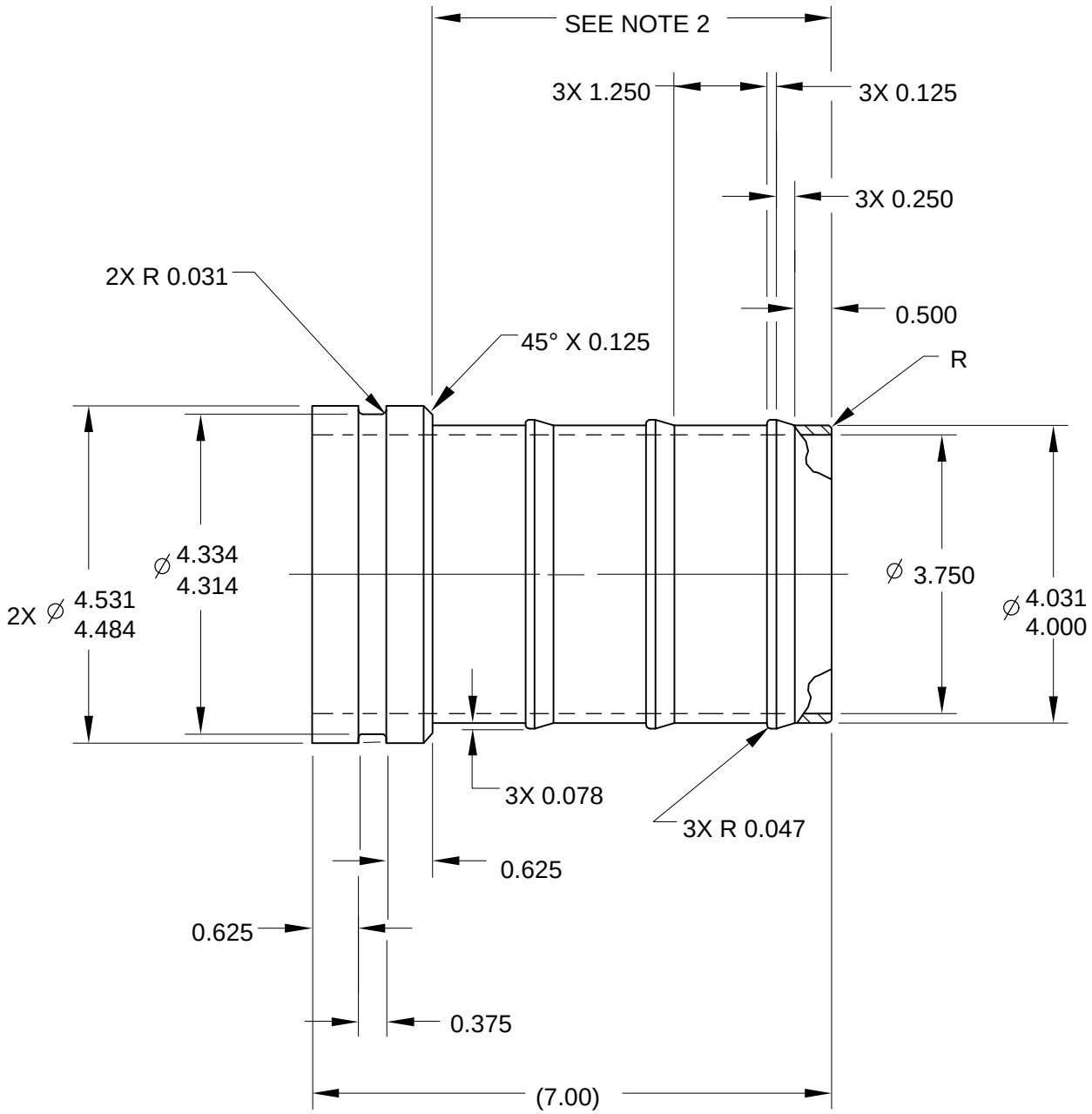
6.5.1 Dissimilar metals. For its purpose, this document considers metals dissimilar when two specimens in contact with each other promote accelerated galvanic corrosion (see 3.10.1.1).

6.5.2 Recovered materials. For the purpose of this document, recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials (see 3.10.2).

6.6 Subject term (key word) listing.

Clamps, hose, 4-inch
Leak repair, hose
Splice, hose
Burst repair, hose
Patch, hose

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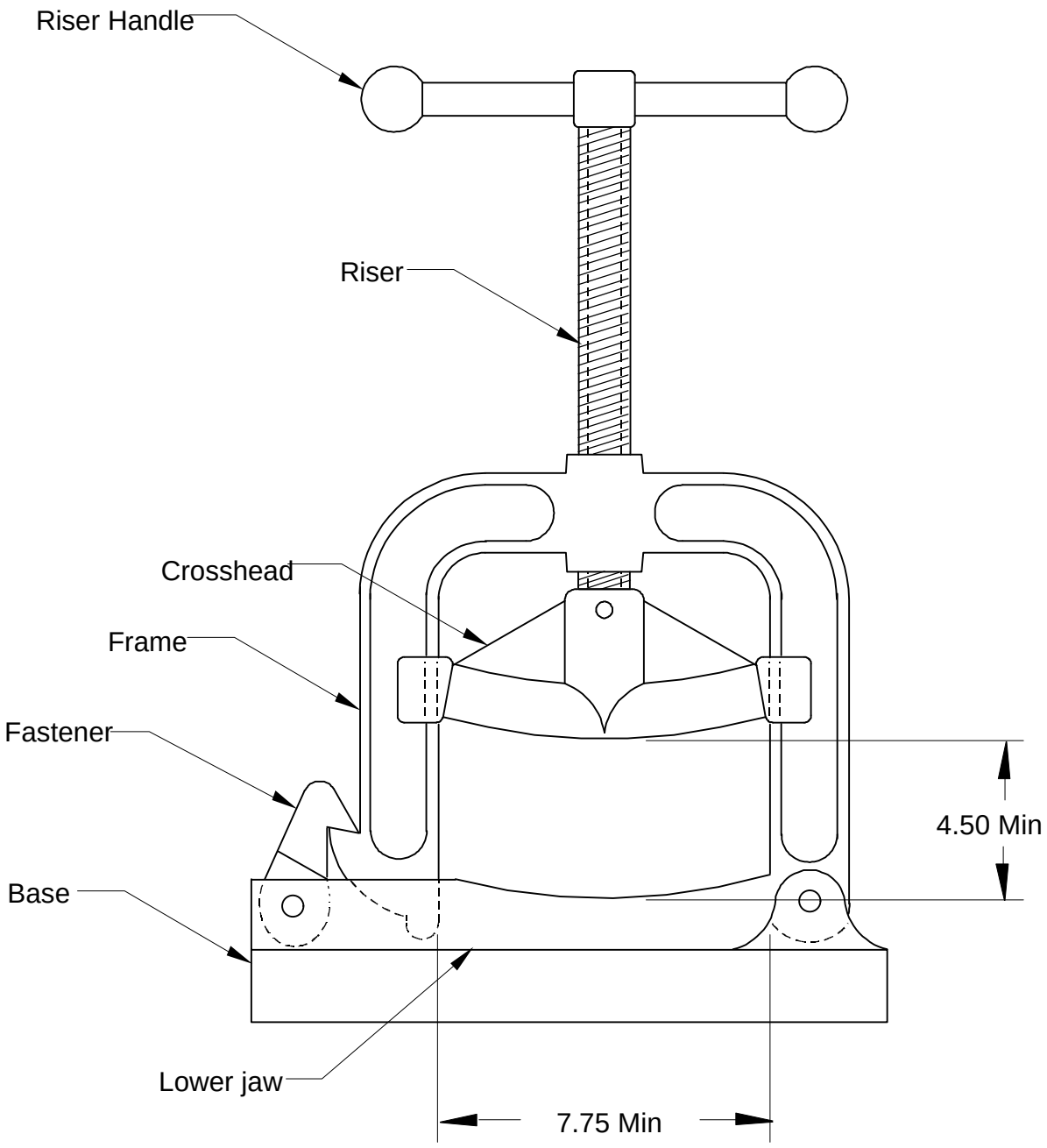


NOTES:

1. Dimensions in inches. Tolerances: .XXX ± .016, ∠ ± 2°
2. Area indicated shall have a matte finish and shall not be sealed after anodizing.

FIGURE 1. Grooved-end to hose adapters – example.

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NOTE: Dimensions are in inches.

FIGURE 2. Hose clamp – example.

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Navy – YD2
Air Force – 99

Review Activities:
Army – DepSO, SM
DLA – CC

Preparing Activity:
Army – AT

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-52343D	2. DOCUMENT DATE (YYMMDD) 980915

3. DOCUMENT TITLE
REPAIR KIT, HOSELINE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME USA TACOM, AMSTA-TR-D/210 (IPDS)	b. TELEPHONE (Including Area Code) (1) Commercial (810) 574-4126 (2) DSN 786-4126
c. ADDRESS (Include Zip Code) US Army Tank-automotive and Armaments Command Attn: AMSTA-TR-D/210 Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 681-9340 DSN 761-9340

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